INCH-POUND
MIL-DTL-16377/36A(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/36(SHIPS)
30 November 1970

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, 21 CANDLEPOWER OR 25 WATTS, 115 VOLTS, WATERTIGHT, SYMBOLS 95.1, 96.1, 97.1, 98.1, 95.1SSL, 96.1SSL, 97.1SSL, AND 98.1SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 7 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

AMSC N/A FSC 6210

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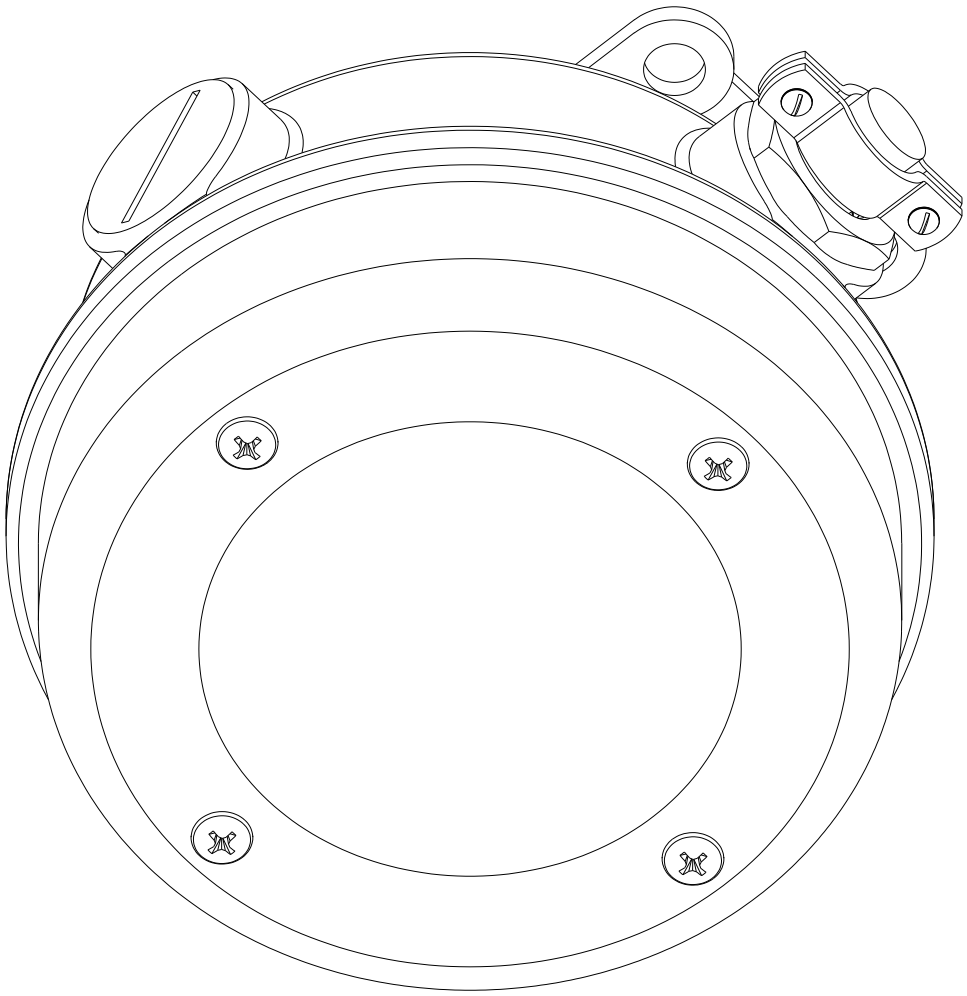


FIGURE 1. Lights, incandescent, symbols 95.1, 96.1, 97.1, and 98.1.

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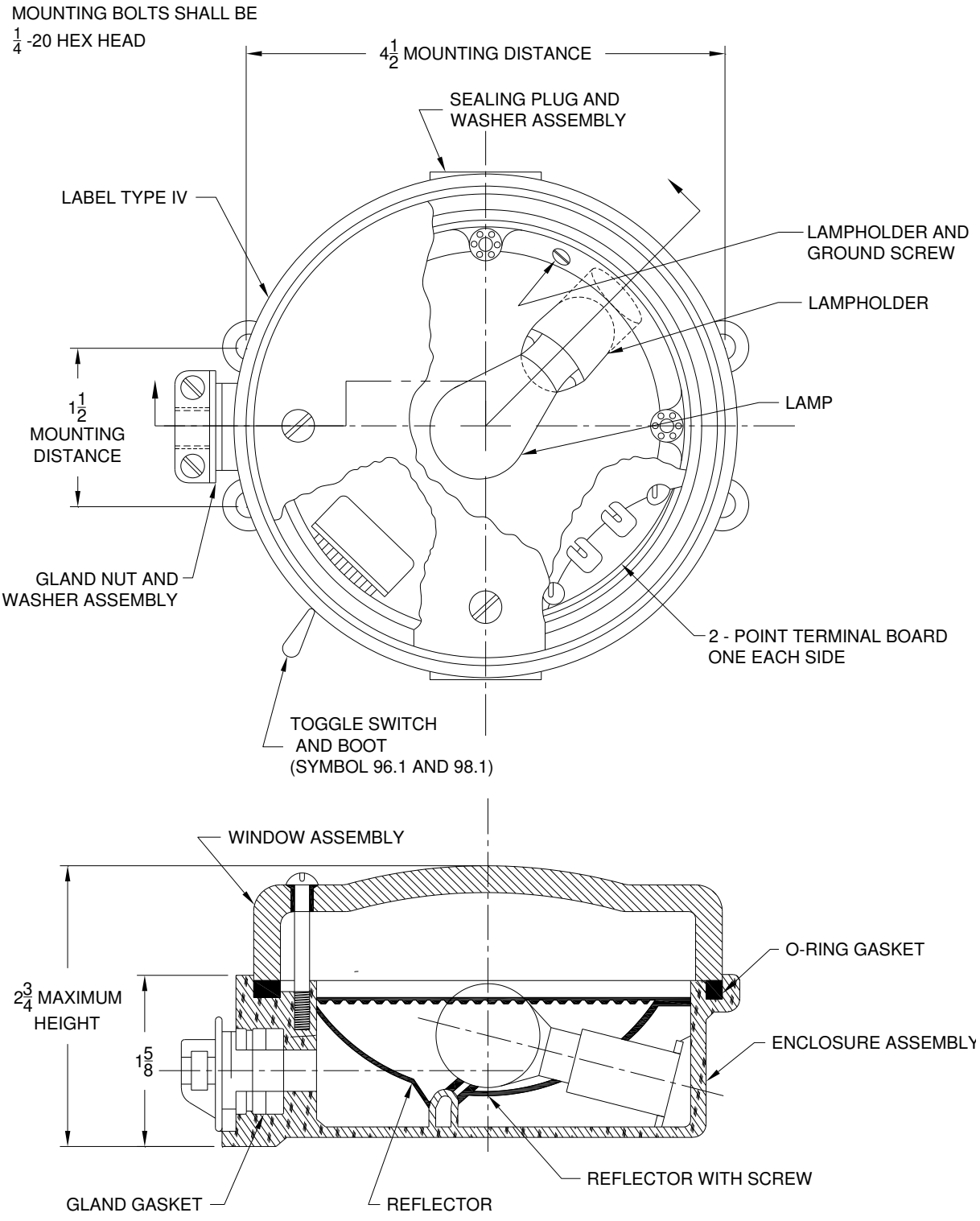


FIGURE 2. Typical view of symbols 95.1, 96.1, 97.1, and 98.1.

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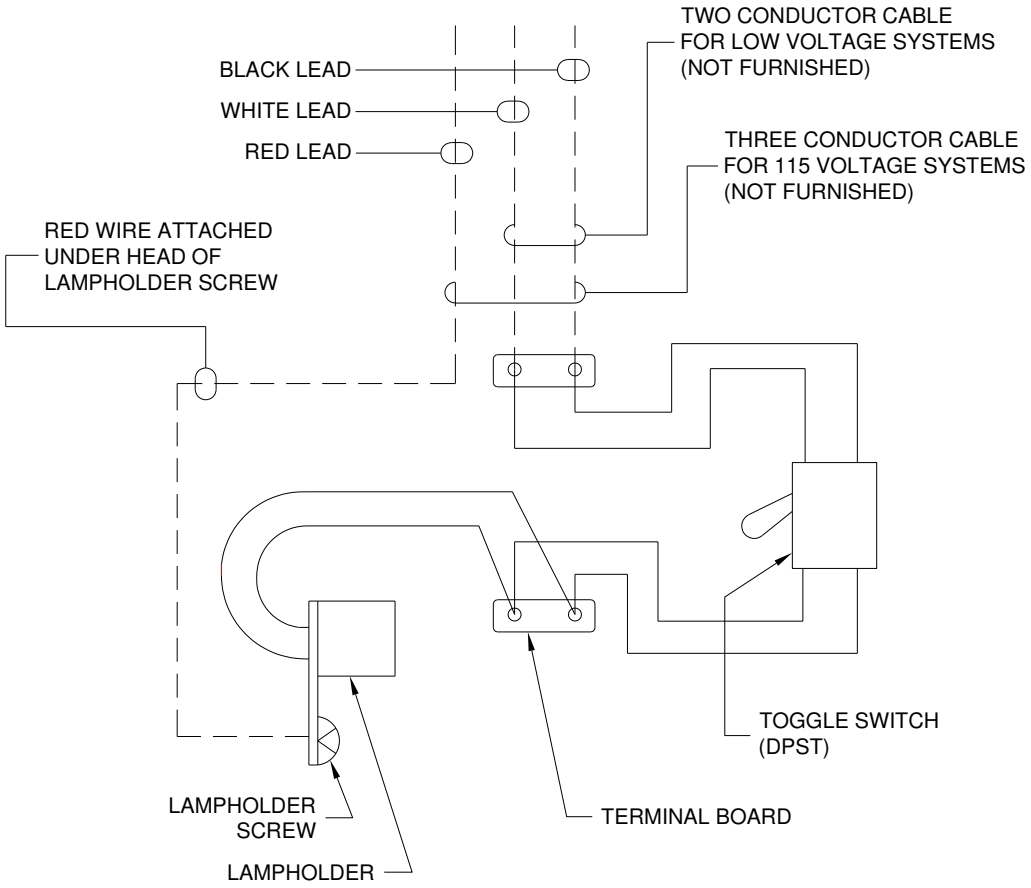


FIGURE 3. Wiring diagram for symbols 96.1 and 98.1.

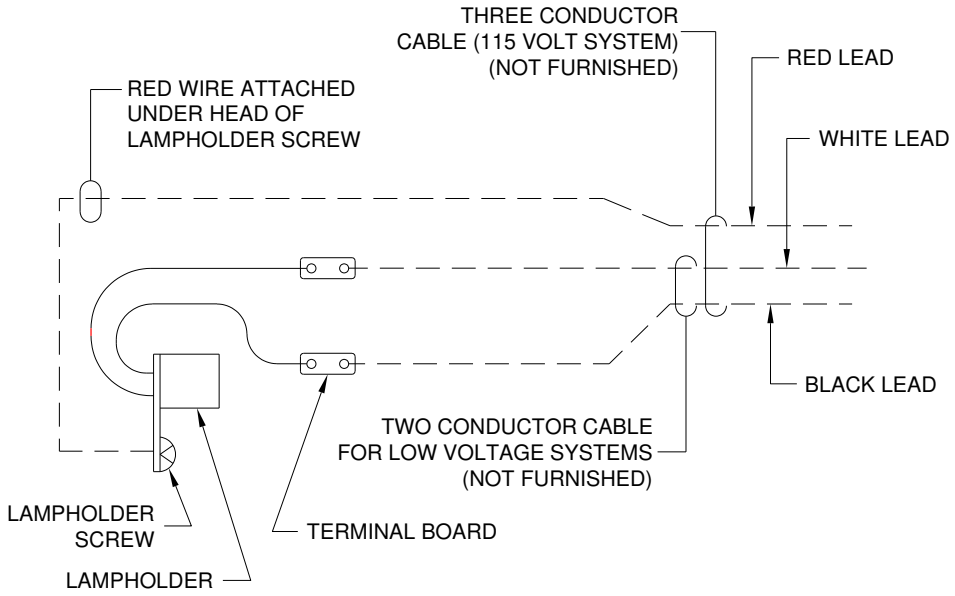


FIGURE 4. Wiring diagram for symbols 95.1 and 97.1.

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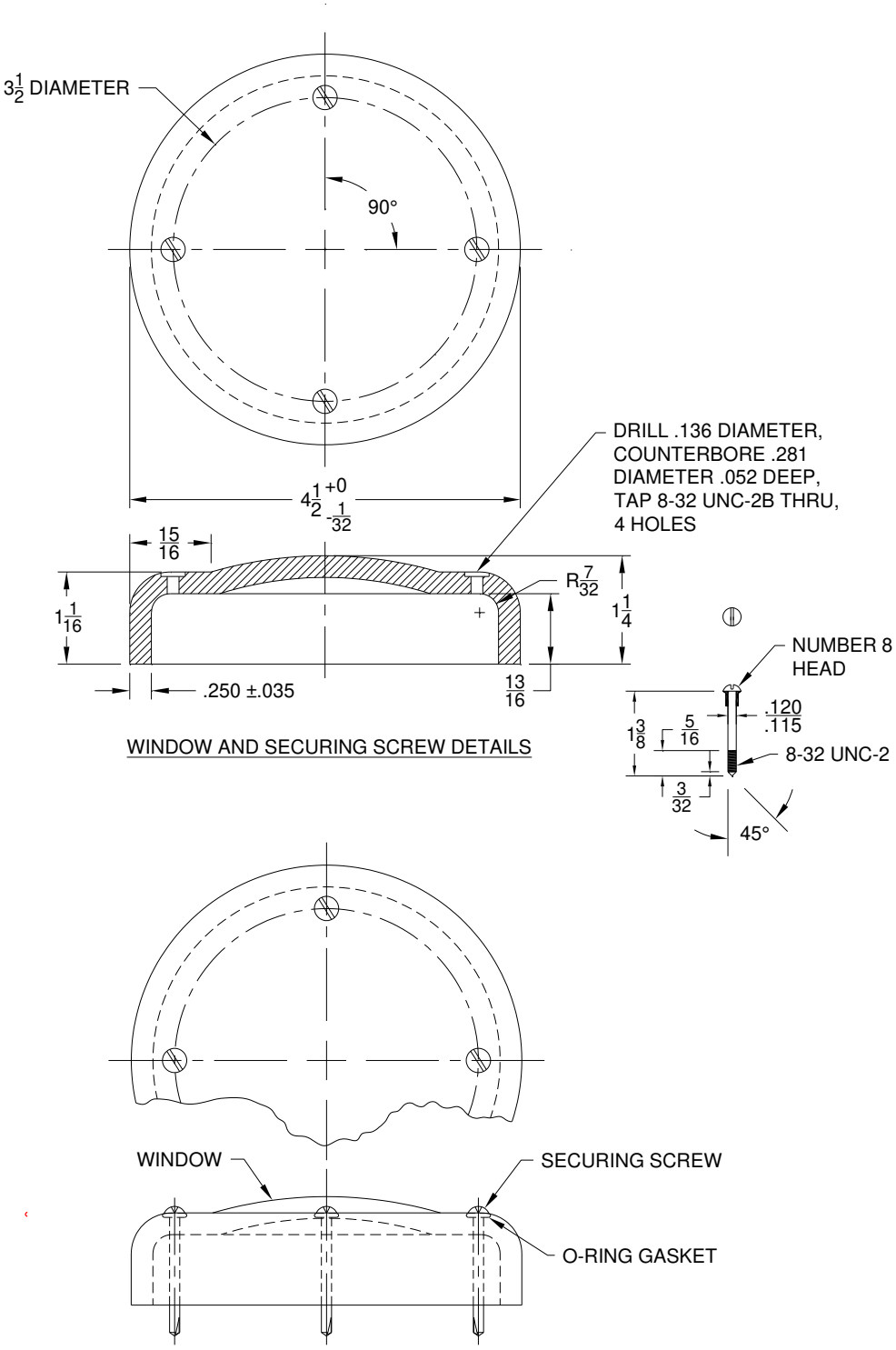
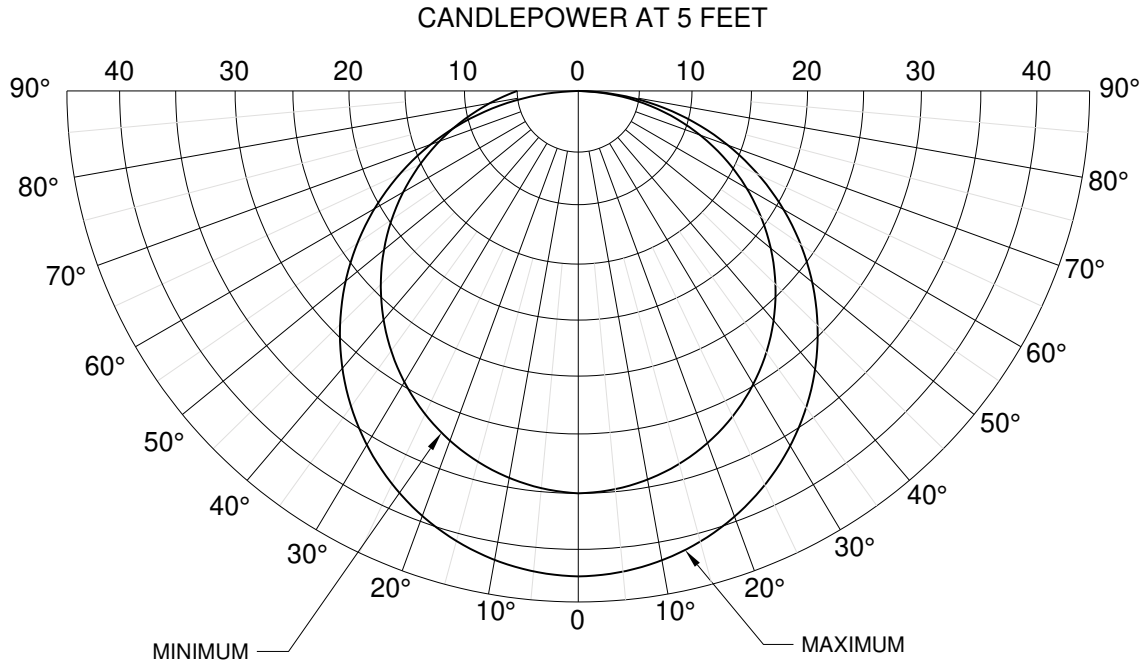


FIGURE 5. Window assembly, diffusing.

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Photometric test data:

Distance: 5 feet

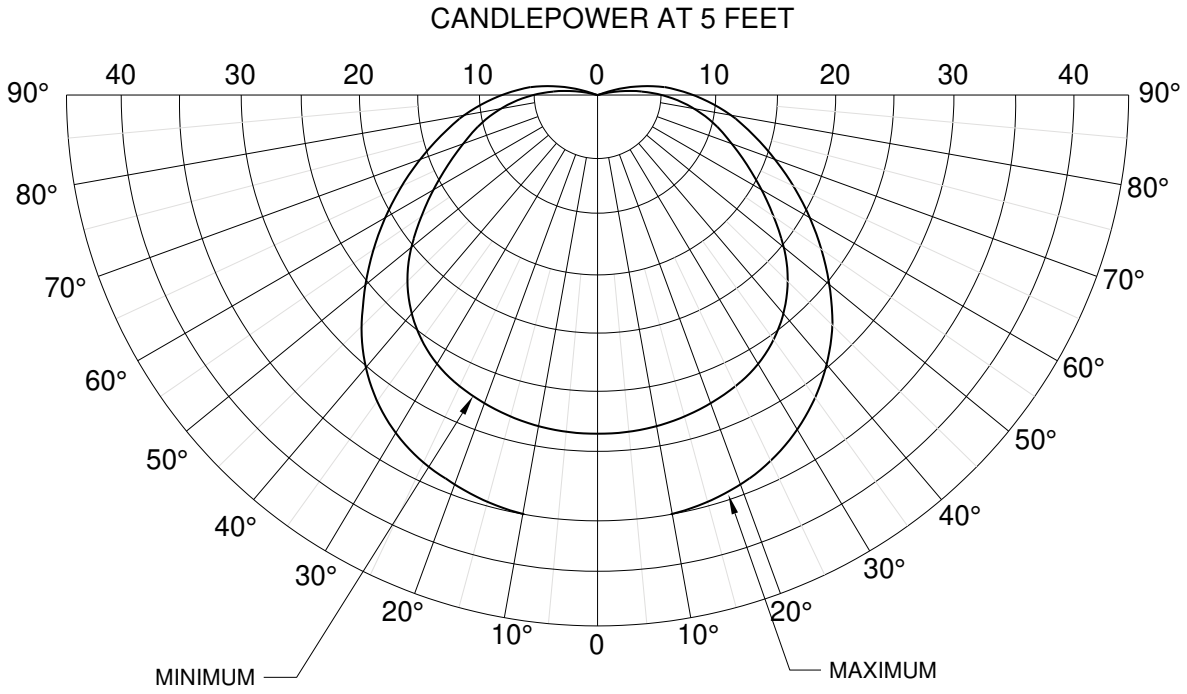
Voltage: 28 VDC

Temperature: 78 °F

Test shall be conducted with one 21-candlepower lamp conforming to industry number 1204 S-8.

FIGURE 6. Candlepower distribution curves for symbols 95.1 and 96.1 (for 21-candlepower lamp).

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Photometric test data:

Distance: 5 feet

Voltage: 115 VAC

Temperature: 78 °F

Test shall be conducted with one 25-watt lamp conforming to A-A-50443.

FIGURE 7. Candlepower distribution curves for symbols 95.1 and 96.1 (for 25-watt lamp).

INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type:	II
Class:	2

SYMBOL NUMBERS

95.1:	Light, with (white diffusing) window.
96.1:	Light, with (white diffusing) window and double pole single through (DPST) switch.
97.1:	Light, with (red diffusing) window.
98.1:	Light, with (red diffusing) window and DPST switch.

DESIGN

Electrical rating:	25 watts, 120 volts AC (VAC) or 28 volts DC (VDC).
Dimensions and configuration:	See figures 1, 2, and 5.
Enclosure assembly:	See figures 1 and 2. All metal parts shall be aluminum, brass, or steel. If steel is used, it shall be protected against corrosion.

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Gaskets:	O-ring: Neoprene rubber, 65-75 durometer hardness, nominal size 3/16 inch wide by 4⅛ inside diameter (id) by 4½ outside diameter (od). Gland gasket: Neoprene sponge, 13/32-inch id by 25/32 inch od by 3/16 inch thick.
Window assemblies:	See figure 5. Material shall be as follows: (a) Windows: (1) Symbols 95.1 and 96.1: White translucent, heat-resistant cast acrylic plastic sheet, ¼ inch thick; type WTLS-4; MIL-DTL-24191, grade G. (2) Symbols 97.1 and 98.1: Red translucent, heat-resistant cast acrylic plastic sheet; ¼ inch thick; MIL-DTL-24191, type RTLS-1. (b) Securing screws: Brass, nickel-plated, or corrosion-resisting steel. (c) O-ring gaskets: SAE AS28775. (d) Coating: Anti-static coating shall be applied to all surfaces of the windows.
Reflector:	Aluminum, white enamel finish.
Lampholder:	UL 496, Supplement SB, bayonet candelabra.
Terminal boards:	Two-point type with laminated phenolic base, commercial design.
Toggle switch for symbols 96.1 and 98.1:	DPST, MS16569-1.
Boot for toggle switch:	Neoprene rubber with integral hexagon nut, enclosed or through type in accordance with MIL-DTL-5423.
Gland nut and washer assembly:	One assembly shall be installed in the light. Three additional assemblies shall be furnished and packaged with the light to provide for additional branch lighting circuits.
Sealing plug and washer assembly:	Three shall be installed in the light as shown on figure 2.
Distribution curve:	Candlepower (see figures 6 and 7).
Lamp:	Light shall be designed to accommodate a 21-candlepower lamp covered by industry numbers 1130, 1142, and 1204 and a 25-watt lamp conforming to A-A-50443. (Lamp not furnished.)
Weight:	1 pound, 5 ounces maximum.
Instruction sheet:	Printed instructions for proper installation of cable and packing arrangements shall be furnished with each light.
Labels:	Labels shall be furnished in accordance with MIL-DTL-16377. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be: "USE ONLY THE FOLLOWING LAMPS: 28 VOLTS – 21 C.P."

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type:	III
Class:	2

SYMBOL NUMBERS

95.1SSL:	Light, white.
96.1SSL:	Light, white, with DPST switch.
97.1SSL:	Light, red.
98.1SSL:	Light, red, with DPST switch.

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DESIGN

Electrical ratings:	12.5 watts maximum, 120 VAC, 60 hertz 12.5 watts, 28 VDC.
Dimensions and configuration:	See figure 2.
Photometric data:	See figure 6 and 3.7.1 of MIL-DTL-16377.
28-VDC interface:	When powered by 28 VDC, the fixture shall utilize ungrounded electric power in accordance with MIL-STD-704 and shall be in accordance with the utilization equipment requirements of MIL-STD-704 for a 28-VDC system. The required performance of the light fixture during the various operating conditions described in MIL-STD-704 is listed in table I. There shall be no bond between any exposed metal part of the fixture and the negative incoming power lead of the 28-VDC system.
Controls:	95.1SSL and 97.1SSL shall not include a control switch; 96.1SSL and 98.1SSL shall include a control switch.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Mounting dimensions:	See figure 6.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Weight:	1 pound, 5 ounces maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.

TABLE I. Requirements for 28-VDC interface.

Operating Condition	Fixture Performance
Normal	100 percent full light output throughout voltage range.
Abnormal	Light output may flicker or go to zero, but the fixture will not suffer permanent damage. Output shall return to full within 5 seconds of normal power being restored.
Emergency	Light output may flicker or be at a reduced output but not at zero percent output. No permanent damage shall be suffered. Output shall return to full within 5 seconds of normal power being restored.
Starting	As specified by acquisition activity.
Transfer	As specified by acquisition activity.
Power failure	No output. No permanent damage shall be suffered. Output shall return to full within 5 seconds of normal power being restored.

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EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for type II and table IV for type III.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

(a) Assignment of military part numbers and general information is shown in table V.

(b) Retaining ring and red and blue window (filter) shall not be installed in the light. They shall be furnished with the light in a bag.

TABLE II. First article and comparison inspection for complete type II fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Grounding circuit	3.6.13.3	4.8.11	Only for grounded incoming cables
Optical uniformity	3.7.4.1.1	4.4	---
Light output	3.7.1	4.8.6.1	Symbols 95.1 and 96.1 only Candlepower
Color and luminous transmittance	3.7.3.6	4.8.6.3	Symbols 97.1 and 98.1 only
Magnetic permeability	3.4.8	4.8.16	---
Continuity of grounding	3.6.13.4	4.8.12	---

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbols 95.1 and 96.1 only Candlepower
Color and luminous transmittance	3.7.3.6	4.8.6.3	Symbols 97.1 and 98.1 only

TABLE IV. First article and comparison inspection for complete type III fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	See table VI for power interface test requirements with 28-VDC power
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14.4	1/ Watertight
Grounding circuit	3.6.13.3	4.8.11	Only for grounded incoming cables
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14.4	1/ Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	1/
Photometrics	3.7.1	4.8.6	---
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---

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TABLE IV. First article and comparison inspection for complete type III fixtures – Continued.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
NOTE: 1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.			

TABLE V. Part numbers and general information.

Military Part Number M16377/36-	Item Name	National Stock Number	Supersedes Part Number
95.1	Complete fixture, symbol 95.1, consisting of body without switch and white window	6210-00-295-2192 6210-00-548-0428	MS17159-95 MS17159-99
96.1	Complete fixture, symbol 96.1, consisting of body with switch and white window	6210-00-295-2865 6210-00-548-0428	MS17159-96
97.1	Complete fixture, symbol 97.1, consisting of body without switch and red window	6210-00-295-2192 6210-00-399-7028	MS17159-97
98.1	Complete fixture, symbol 98.1, consisting of body with switch and red window	6210-00-295-2865 6210-00-399-7028	MS17159-98
95.1SSL	Complete fixture, symbol 95.1SSL	---	---
96.1SSL	Complete fixture, symbol 96.1SSL	---	---
97.1SSL	Complete fixture, symbol 97.1SSL	---	---
98.1SSL	Complete fixture, symbol 98.1SSL	---	---
001	Body with switch	6210-00-295-2865	---
002	Body without switch	6210-00-295-2192	---
003	White window	6210-00-548-0428	MS16788-3
004	Red window	6210-00-399-7028	MS16788-4

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TABLE VI. MIL-HDBK-704-8 interface test requirements for 28-VDC input power.

MIL-HDBK-704-8 Test Method	Notes
Method LDC101 - Load Measurements	In-rush current: N/A Load: 12.5W Current Distortion Factor Current Spectrum
Method LDC102 - Steady State Limits for Voltage	---
Method LDC103 - Voltage Distortion Spectrum	---
Method LDC104 - Total Ripple	---
Method LDC105 - Normal Voltage Transients	---
Method LDC201 - Power Interrupt	---
Method LDC301 - Abnormal Steady State Limits for Voltage	---
Method LDC302 - Abnormal Voltage Transients (Overvoltage/Undervoltage)	---
Method LDC401 - Emergency Limits for Voltage	---
Method LDC501 - Starting Voltage Transients	---
Method LDC601 - Power Failure	---
Method LDC602 - Polarity Reversal	---

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-014)

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